Holding Tanks and Agitators

UBC CHBE capstone · Montbretin A production · PFD 400: T-401/M-401, T-402/M-402

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Given

	T-401 / M-401 (centrifugation holding tank)	**T-402 / M-402** (TFF holding tank)
Maximum liquid volume, $V_{\max}$	$16\,\mathrm{m}^3$ (harvest)	$4.8\,\mathrm{m}^3$ (TFF feed)
Minimum liquid volume, $V_{\min}$	– (constant during wash)	$1.59\,\mathrm{m}^3$ (end of concentration mode)
Design tank volume, V	$18\,\mathrm{m}^3$	$5.2\,\mathrm{m}^3$
Liquid density, ρ	$1002\,\mathrm{kg/m}^3$ (broth)	$1003.2\,\mathrm{kg/m}^3$ (lysate/retentate)
Viscosity, μ	$0.001\,\mathrm{Pa\,s}$ (water)	
Geometry	$H_{\mathrm{tank}}/D_{\mathrm{tank}} = 2, \quad D_{\mathrm{tank}}/D_i = 3$	
Impeller tip speed, v_i	$4\,\mathrm{m/s}$	
Power number, N_P	3 (basis: turbulent mixing, $\mathrm{Re}_i > 10^4$)	

Find

Tank diameter and height; minimum and maximum liquid heights; agitator shaft length, impeller diameter, speed, Reynolds number and power.

Equations

$$V = \pi \left(\frac{D_{\mathrm{tank}}}{2}\right)^2 (2D_{\mathrm{tank}}) = \frac{\pi D_{\mathrm{tank}}^3}{2} \;\Rightarrow\; D_{\mathrm{tank}} = \sqrt[3]{\frac{2V}{\pi}} \tag{1}$$

$$h_{\mathrm{liq}} = \frac{V_{\mathrm{liq}}}{V}\, H_{\mathrm{tank}} \tag{2}$$

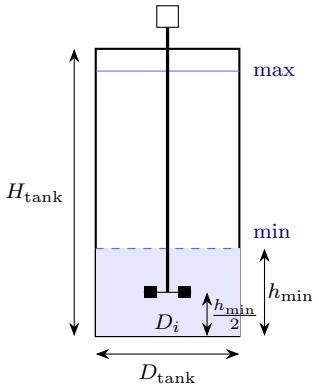
$$L_{\mathrm{shaft}}^{\text{T-402}} = H_{\mathrm{tank}} - \frac{h_{\mathrm{liq,min}}}{2} \tag{3}$$

$$L_{\mathrm{shaft}}^{\text{T-401}} = H_{\mathrm{tank}} - D_i \tag{4}$$

$$D_i = \frac{D_{\mathrm{tank}}}{3}, \qquad N_i = \frac{v_i}{\pi D_i} \tag{5}$$

$$\mathrm{Re}_i = \frac{\rho\, N_i D_i^2}{\mu} \tag{6}$$

$$N_P = \frac{P}{\rho\, N_i^3 D_i^5} \;\Rightarrow\; P = N_P\, \rho\, N_i^3 D_i^5 \tag{7}$$



Assumptions

- Flat-bottomed cylindrical tanks with $H/D = 2$ and $D_{\mathrm{tank}}/D_i = 3$ (course heuristics).
- Design volumes above the maximum liquid volume leave a clearance margin: $5.2\,\mathrm{m}^3$ for $4.8\,\mathrm{m}^3$, $18\,\mathrm{m}^3$ for $16\,\mathrm{m}^3$.
- T-402: impeller at half the minimum liquid height, so it stays submerged as the level falls during filtration.

- T-401: working volume is constant ($16\,\mathrm{m}^3$) during the wash, so there is no falling level to protect; the impeller sits one impeller diameter off the floor ($H_i/D_i = 1$).
- Tip speed of about $4\,\mathrm{m/s}$ for blending.
- Fully turbulent mixing in a baffled tank, so N_P is constant (chart basis: 4 baffles, $W_b/D_T = 1/12$).
- Water viscosity; density constant over the operation.

Worked example

1. T-402 (TFF holding tank)

The maximum volume is $4.8\,\mathrm{m}^3$, so a $5.2\,\mathrm{m}^3$ basis was used for clearance margin.

$$D_{\text{tank}} = \sqrt[3]{\frac{2\,(5.2\,\mathrm{m}^3)}{\pi}} = \boxed{1.490\,\mathrm{m}}, \qquad H_{\text{tank}} = 2\,(1.490\,\mathrm{m}) = \boxed{2.981\,\mathrm{m}}$$

$$h_{\text{liq,min}} = \frac{1.59\,\mathrm{m}^3}{5.2\,\mathrm{m}^3}\,(2.981\,\mathrm{m}) = \boxed{0.911\,\mathrm{m}}, \qquad h_{\text{liq,max}} = \frac{4.8\,\mathrm{m}^3}{5.2\,\mathrm{m}^3}\,(2.981\,\mathrm{m}) = \boxed{2.751\,\mathrm{m}}$$

2. M-402 (T-402 agitator)

$$L_{\text{shaft}} = 2.981\,\mathrm{m} - \frac{0.911\,\mathrm{m}}{2} = \boxed{2.525\,\mathrm{m}}, \qquad D_i = \frac{1.490\,\mathrm{m}}{3} = \boxed{0.497\,\mathrm{m}}$$

$$N_i = \frac{v_i}{\pi D_i} = \frac{4\,\mathrm{m/s}}{\pi\,(0.497\,\mathrm{m})} = \boxed{2.563\,\mathrm{rev/s}}\ (154\,\mathrm{rpm})$$

$$\mathrm{Re}_i = \frac{(1003.2\,\mathrm{kg/m}^3)(2.563/\mathrm{s})(0.497\,\mathrm{m})^2}{0.001\,\mathrm{Pa\,s}} = 6.35 \times 10^5 \ \Rightarrow\ \text{turbulent},\ N_P = 3$$

$$P = (3)(1003.2\,\mathrm{kg/m}^3)(2.563/\mathrm{s})^3(0.497\,\mathrm{m})^5 \times \frac{1\,\mathrm{kW}}{1000\,\mathrm{W}} = \boxed{1.53\,\mathrm{kW}}$$

3. T-401 (centrifugation holding tank) and M-401

Same method. The maximum volume is $16\,\mathrm{m}^3$, so an $18\,\mathrm{m}^3$ basis was used; the stream-table workbook takes the diameter as its input, still set from this working volume.

$$D_{\text{tank}} = \sqrt[3]{\frac{2\,(18\,\mathrm{m}^3)}{\pi}} = \boxed{2.255\,\mathrm{m}}, \qquad H_{\text{tank}} = 2\,(2.255\,\mathrm{m}) = \boxed{4.51\,\mathrm{m}}$$

$$D_i = \frac{2.255\,\mathrm{m}}{3} = 0.752\,\mathrm{m}, \qquad L_{\text{shaft}} = 4.51\,\mathrm{m} - 0.752\,\mathrm{m} = \boxed{3.758\,\mathrm{m}}$$

$$N_i = \frac{4\,\mathrm{m/s}}{\pi\,(0.752\,\mathrm{m})} = 1.694\,\mathrm{rev/s}\ (102\,\mathrm{rpm})$$

$$\mathrm{Re}_i = \frac{(1002)(1.694)(0.752)^2}{0.001} = 9.59 \times 10^5, \qquad P = (3)(1002)(1.694)^3(0.752)^5/1000 = \boxed{3.51\,\mathrm{kW}}$$

Summary

	D_{tank} [m]	H_{tank} [m]	L_{shaft} [m]	D_i [m]	N_i [rev/s]	Re_i	P [kW]
T-402 / M-402	1.490	2.981	2.525	0.497	2.563	6.35×10^5	1.53
T-401 / M-401	2.255	4.51	3.758	0.752	1.694	9.59×10^5	3.51

Check

- **Volume:** $\frac{\pi}{4}(1.490)^2(2.981) = 5.20\,\mathrm{m}^3$. The team workbook computed D_{tank} as $(2V/\pi)/3 = 1.10\,\mathrm{m}$ (Excel applies `^1` before `/3`); that tank holds only $2.11\,\mathrm{m}^3$, less than the $4.8\,\mathrm{m}^3$ batch. Corrected here to $1.490\,\mathrm{m}$.
- **Impeller placement:** at minimum level the impeller sits $0.456\,\mathrm{m}$ off the floor, close to the common $D_{\mathrm{tank}}/3 = 0.497\,\mathrm{m}$ clearance, with $0.456\,\mathrm{m}$ ($\approx 0.9\,D_i$) of liquid above it.
- **Turbulence:** both $\mathrm{Re}_i > 10^4$, so a constant N_P applies.